

Daily Practice AP Calculus Unit 5

Topics: Analytical Applications of Differentiation (Extreme Value Theorem, Mean Value Theorem, First and Second Derivative Tests, Concavity)

Section 1: Multiple Choice Questions

No calculator is allowed for these questions.

1. What is the absolute maximum value of $f(x) = x^3 - 3x^2 + 1$ on the closed interval $[-1, 4]$?
(A) -3
(B) 1
(C) 5
(D) 17
2. Let f be the function given by $f(x) = \sin(x)$ on the closed interval $[0, \pi]$. What is the value of c in the open interval $(0, \pi)$ that satisfies the conclusion of the Mean Value Theorem?
(A) $\frac{\pi}{4}$
(B) $\frac{\pi}{2}$
(C) $\frac{3\pi}{4}$
(D) The Mean Value Theorem does not apply.
3. The first derivative of the function f is given by $f'(x) = 4x^3 - 12x^2$. What are the x -coordinates of the relative minima of f ?
(A) $x = 0$ only
(B) $x = 3$ only
(C) $x = 0$ and $x = 3$
(D) The function has no relative minima.
4. On which of the following open intervals is the graph of $f(x) = x^4 - 4x^3$ both decreasing and concave up?
(A) $(-\infty, 0)$ and $(2, 3)$
(B) $(0, 2)$

- (C) $(2, 3)$
- (D) $(3, \infty)$
5. Let f be a function whose second derivative is given by $f''(x) = x(x - 2)^2(x + 3)$. At what values of x does the graph of f have a point of inflection?
- (A) $x = 2$ only
- (B) $x = 0$ and $x = -3$ only
- (C) $x = 0$ and $x = 2$ only
- (D) $x = -3$, $x = 0$, and $x = 2$
6. Let f be a function that is continuous on the closed interval $[-2, 2]$ and differentiable on the open interval $(-2, 2)$. If $f(-2) = 5$ and $f(2) = 5$, which of the following statements MUST be true?
- (A) $f(x) > 0$ for all x in $[-2, 2]$.
- (B) $f'(c) = 0$ for at least one c in the open interval $(-2, 2)$.
- (C) f has a relative maximum in the open interval $(-2, 2)$.
- (D) $f''(x) < 0$ for all x in the open interval $(-2, 2)$.
7. The function f is given by $f(x) = x^{2/3}$. Which of the following statements is true about the critical points of f ?
- (A) $x = 0$ is a critical point because $f'(0) = 0$.
- (B) $x = 0$ is a critical point because $f'(0)$ is undefined and $x = 0$ is in the domain of f .
- (C) There are no critical points because $f'(x) \neq 0$ for all x .
- (D) There are no critical points because the derivative is never undefined on the domain of f .
8. If $f'(x) > 0$ for all real numbers x and $f''(x) < 0$ for all real numbers x , which of the following could be a table of values for f ?
- (A) $f(1) = 2, f(2) = 3, f(3) = 5$
- (B) $f(1) = 2, f(2) = 4, f(3) = 5$
- (C) $f(1) = 5, f(2) = 3, f(3) = 2$
- (D) $f(1) = 5, f(2) = 4, f(3) = 2$

Section 2: Free Response Questions

Show all your work clearly.

FRQ 1. Let f be the function defined by $f(x) = \frac{1}{3}x^3 - x^2 - 3x + 2$.

(a) Find the x -coordinates of all relative extrema of f . Classify each as a relative maximum or relative minimum. Justify your answers.

(b) Find the x -coordinates of all points of inflection of the graph of f . Justify your answer.

(c) Find the absolute maximum and absolute minimum values of f on the closed interval $[-2, 4]$. Show the analysis that leads to your conclusion.

FRQ 2. A continuous and twice-differentiable function $g(x)$ is defined for all real numbers. Selected values of $g(x)$ are given in the table below.

x	$g(x)$
0	1
2	5
4	2
6	8

(a) Must there exist a value c in the open interval $(0, 2)$ such that $g'(c) = 2$? State the theorem you are using and verify its conditions.

(b) Must there exist a value k in the open interval $(2, 6)$ such that $g'(k) = 0$? Justify your answer.

(c) It is given that $g''(x) > 0$ for all x in the open interval $(4, 6)$. Is the average rate of change of g on the interval $[4, 5]$ greater than, less than, or equal to the average rate of change of g on the interval $[5, 6]$? Explain your reasoning.